

concept mapping

- 1) construct a concept map using complete machine learning workflow for predicting house price using suitable M.L work flow.
- 2) construct a concept map demonstrating how classification algorithms are used in medical disease prediction.
e.g:- Diabetes or Heart disease prediction.
- 3) construct a concept map showing how machine learning is applied in credit card fraud detection.
- 4) construct a concept map explaining student performance prediction using the multiple regression algorithm.
- 5) construct a concept map showing how decision tree classification is used for loan approval.

Solutions

- 1) house prediction system.

STEP 1:- Key words

- a) House area
- b) location
- c) Bedroom
- d) House id
- e) price
- f) age

STEP 2:- Relationship table

Step 3: Demonstration of Datasets

House id. (S.no.)	Area (sq ft)	Bedroom	age	location score	Price
1	500 sq ft.	2.	3	8	16 lakhs
2	650 sq ft	3.	6.	9	18 lakhs
3.	250 sq ft.	4.	2.	7	10 lakhs
4.	350 sq ft	2.	4	6	9 lakhs.

Step 4: data flow

House dataset

↳ data cleaning

→ data preprocessing

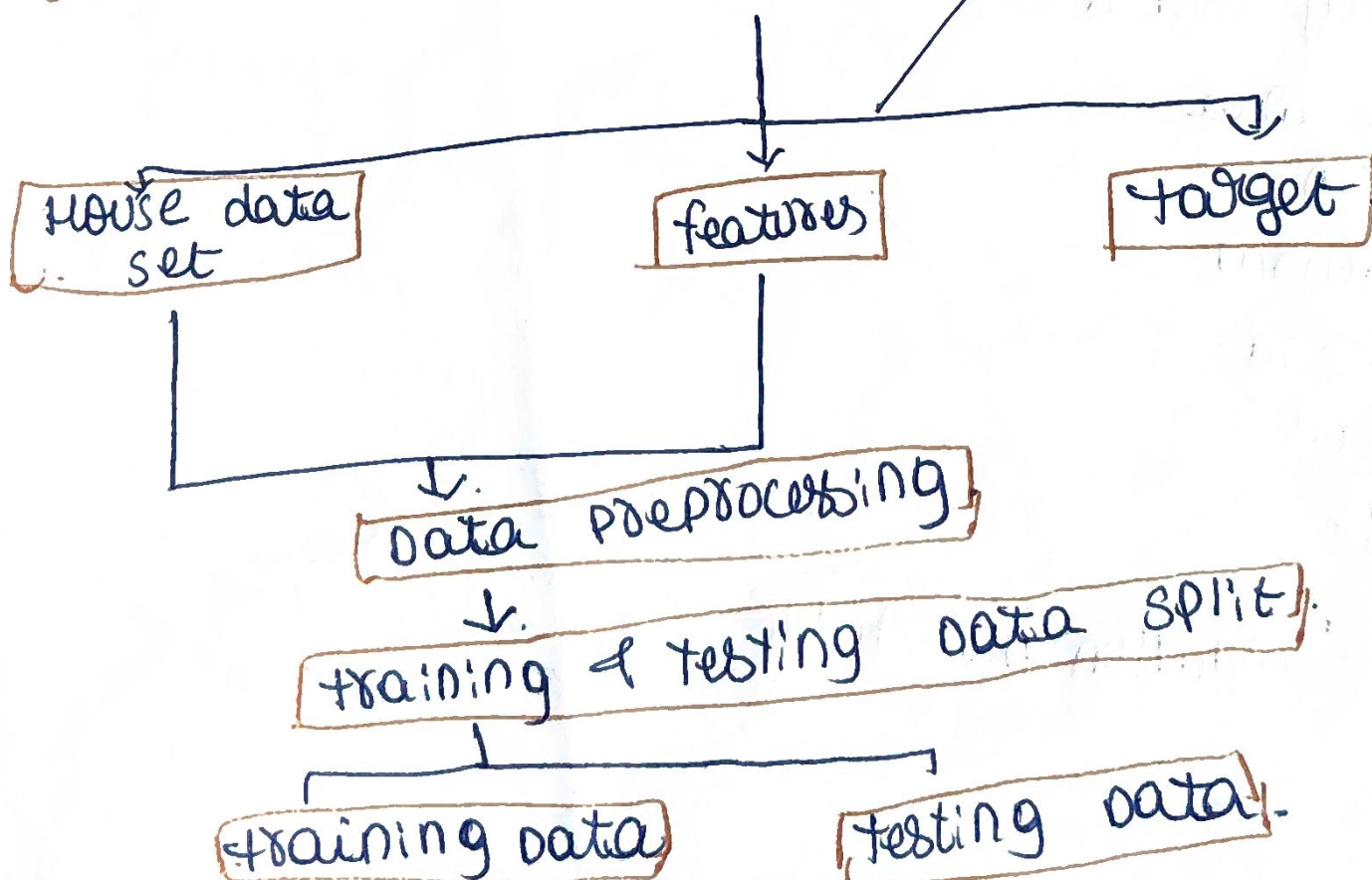
(cleaning, encoding)

training data

testing data.

Step 5:

House price prediction



2) medical disease prediction:-

Step 1:- keywords

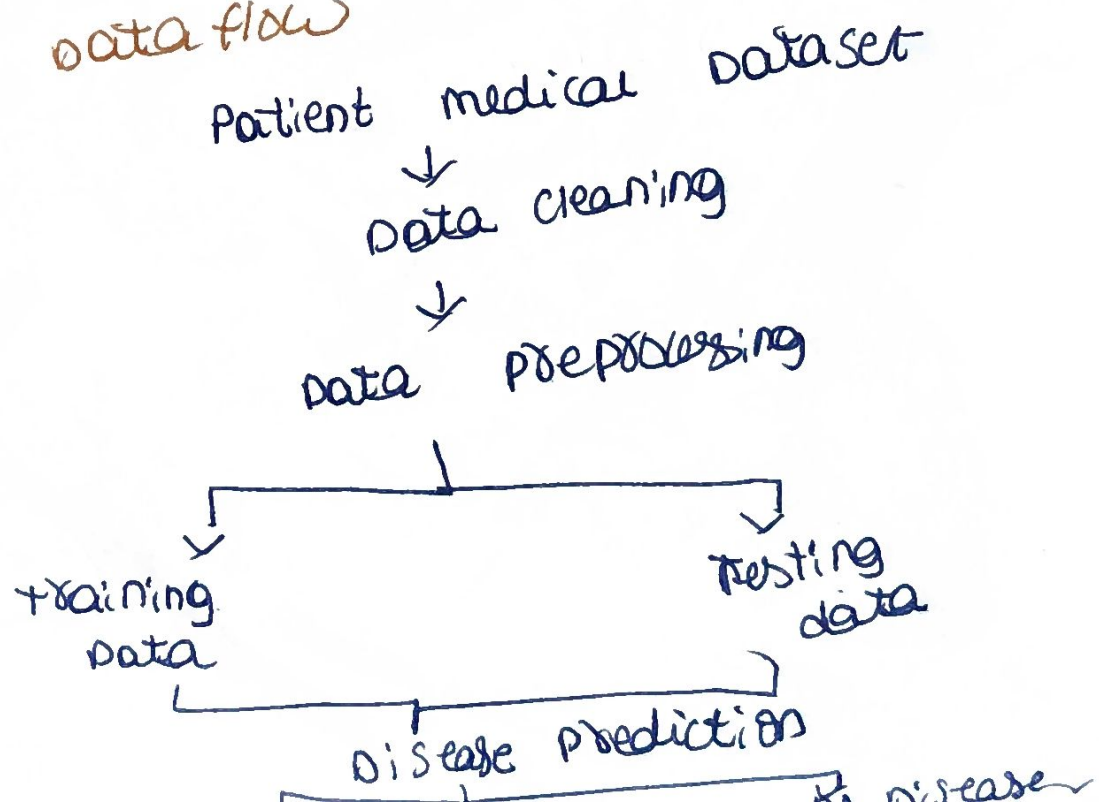
- a) Patient health status-
- b) age
- c) glucose level
- d) blood pressure
- e) BMI.
- f) cholesterol.

Step 2:- demonstration of dataset.

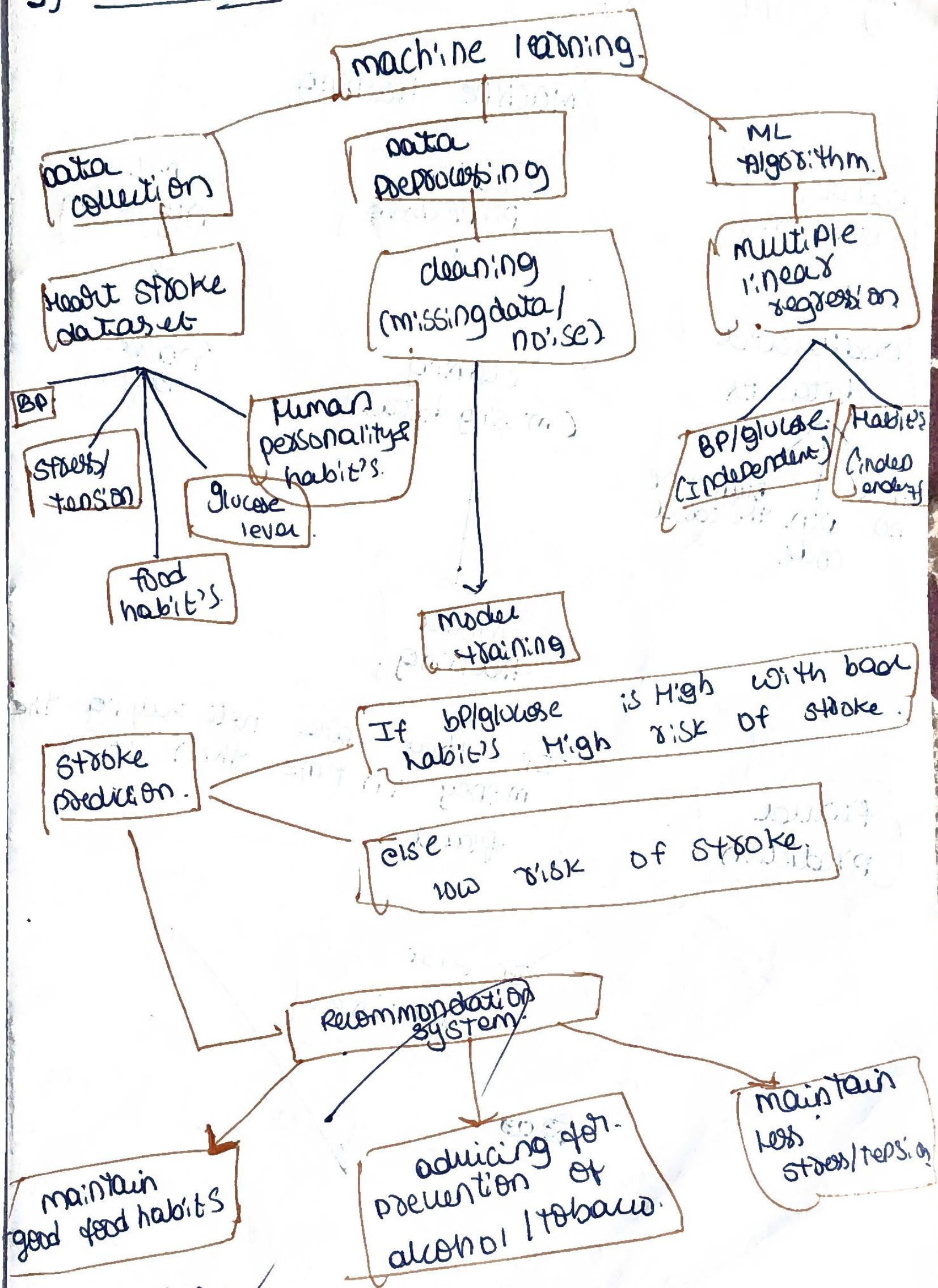
sn	patient ID	age	glucose level	cholesterol	disease.
1	136	45	150	210	yes
2	137	35	95	180	no
3	138	55	180	240	yes
4	139	65	90	260	yes
5	140	30	99	170	no

Step 3:-

data flow



2) Heart disease prediction.



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credit card fraud detection

Keywords:-

- a) credit card.
- b) Transaction data set.
- c) fraudulent transaction.
- d) genuine transaction.
- e) fraud transaction.

Demonstration of dataset

S.No	Transaction ID	Amount	Time	Location	Fraud
1	₹500	1	10:30AM	Chennai	NO
2	₹25,600	2	2:00AM	Mumbai	Yes
3	₹1,200	3	1:00PM	Chennai	NO
4	₹50,000	4	3:30PM	Delhi	Yes.

Dataflow:-

credit card transaction

↓
data cleaning

↓
Data processing

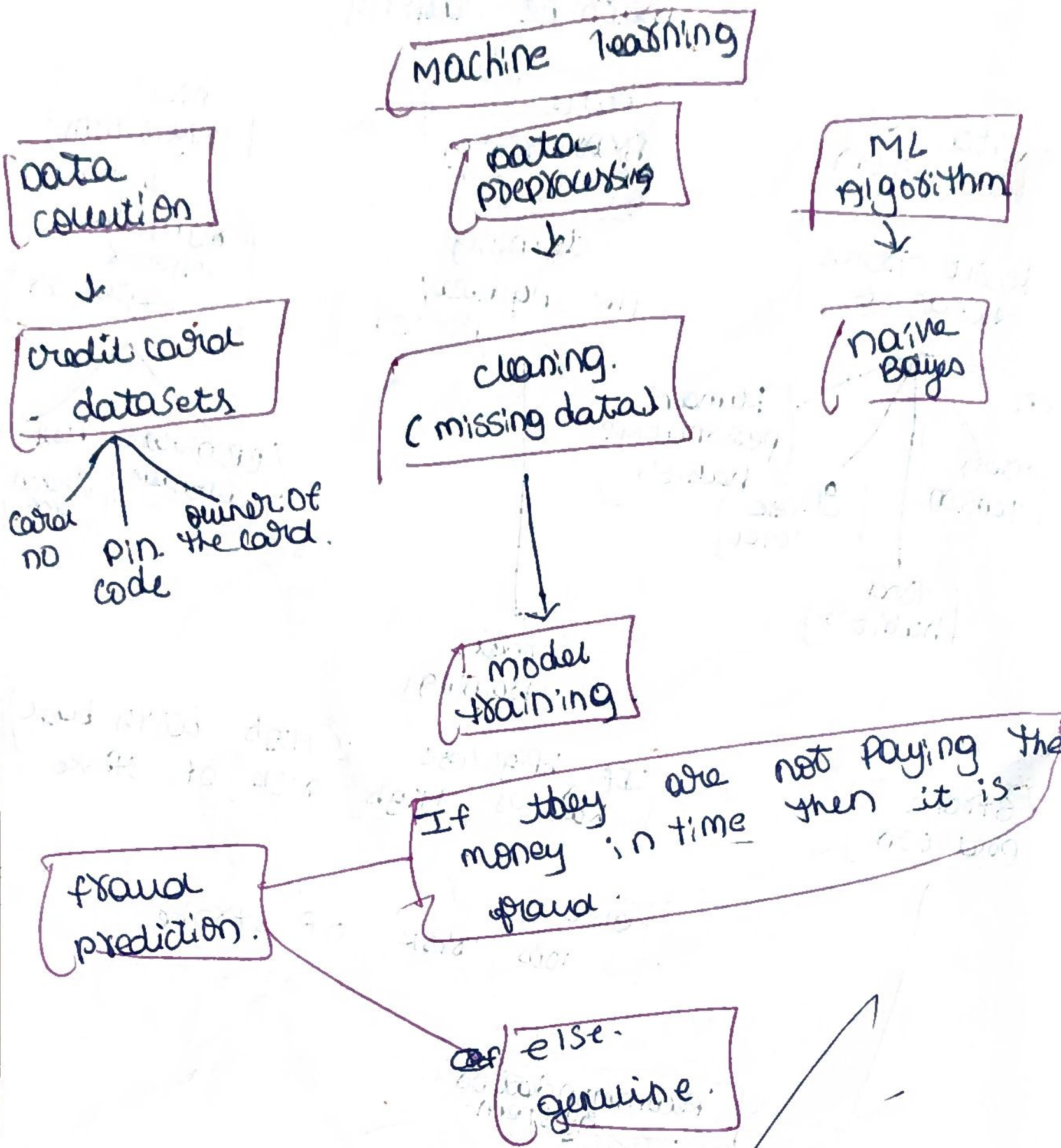
↓
feature selection.

┌──────────┴──────────┐
training data transaction data.

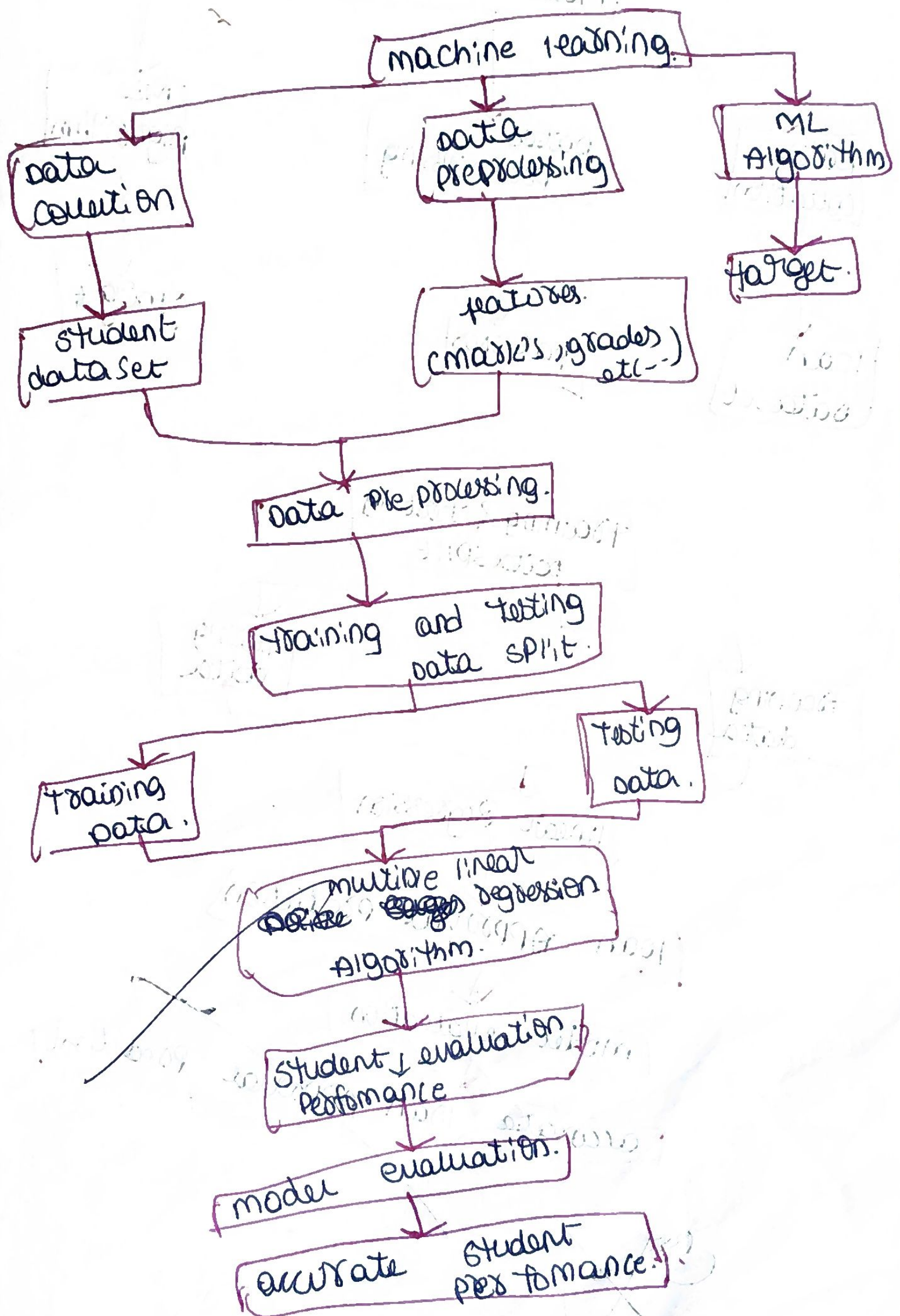
└──────────┬──────────┘
 fraud detection.

┌──────────┴──────────┐
genuine fraud

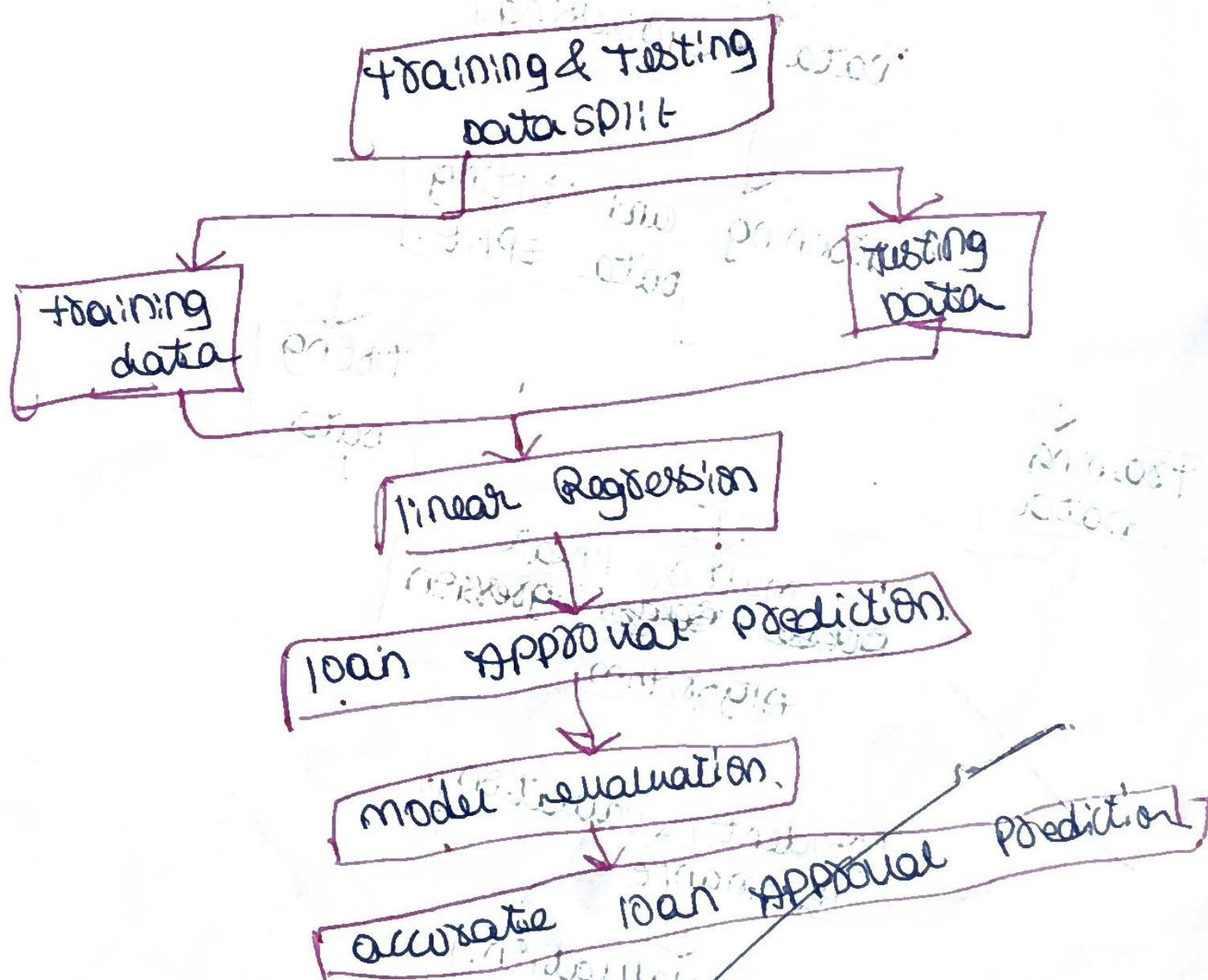
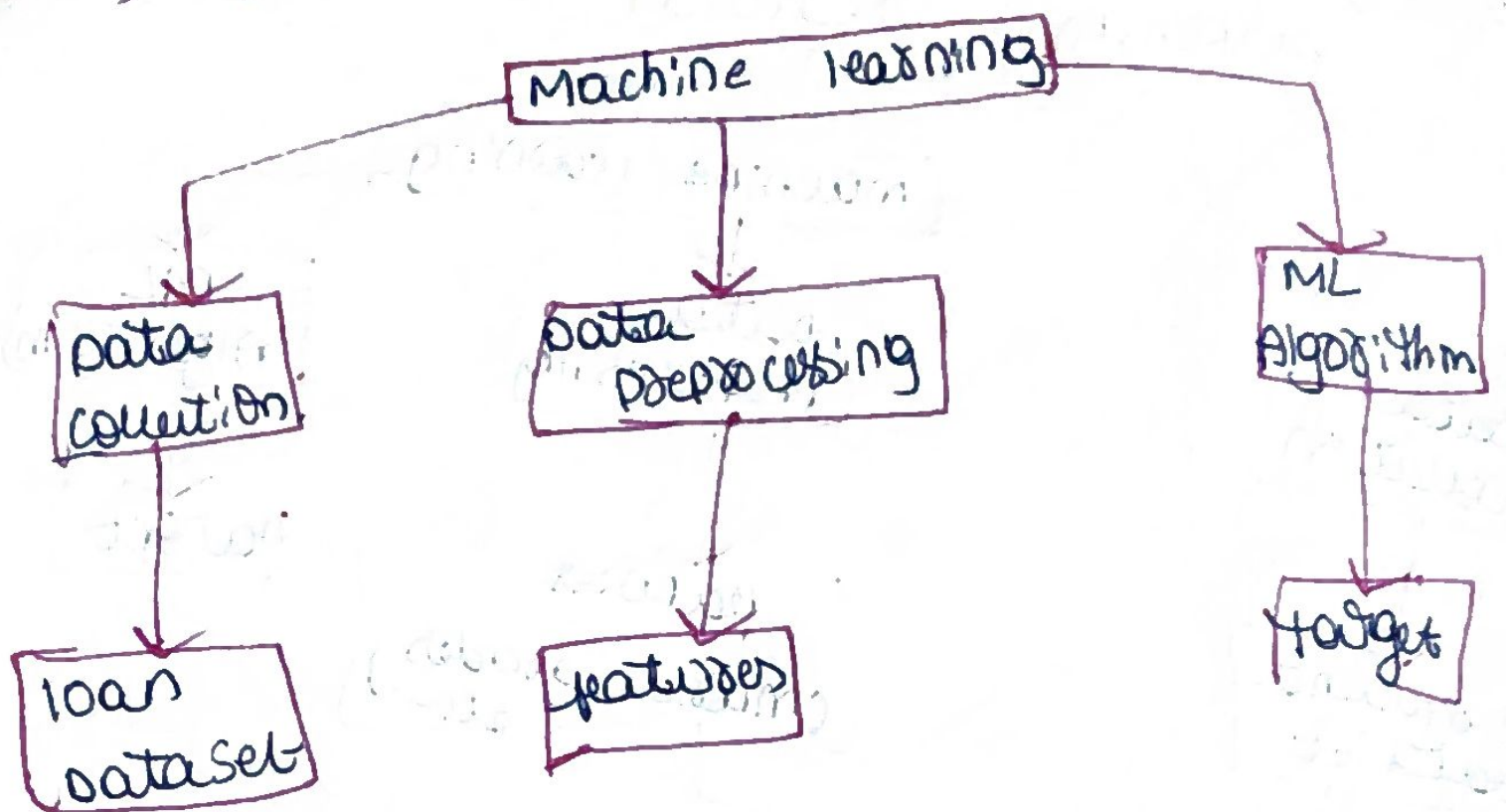
3) Credit card fraud detection



4) Student performance prediction.



5) Loan Approval Prediction.



Signature